Schooling and Earnings Across U.S. States, 2022–2024

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Repository: https://github.com/Jiayi-Eva-Wang/econ3916-midterm-project

Notebook: https://colab.research.google.com/drive/127m-JJzKB1XmGt2Cij1kSByAM6MRh6hK?usp=sharing

Question	How is estimated average schooling among adults aged 25 and over associated with median annual earnings across the 50 U.S. states from 2022 to 2024, after controlling for population size and median age?
Hypothesis (optional)	I expect states with higher estimated average schooling to have higher median annual earnings; Education may be associated with skills valued in the labor market and access to higher-paying occupations. However, differences in industry composition, migration, and local economic conditions may also contribute to this relationship.
Headline finding	Across the 50 U.S. states, higher estimated average schooling is positively associated with median annual earnings from 2022 to 2024. In 2024, a 0.1-year higher estimated state average schooling level is associated with approximately $1,466 higher median earnings, controlling for population size and median age (95% CI: $1,114–$1,818; N = 50). This is a conditional association, not a causal effect.
Data	One row is one state-year (up to 150 observations across the 50 U.S. states, 2022–2024). Source: U.S. Census Bureau, ACS 1-Year Estimates, Tables B15003 (educational attainment), B20004 (median annual earnings among adults aged 25+ with earnings), B01003 (population), and B01002 (median age). Average schooling years are estimated from educational attainment categories.
Method	Separate OLS regressions for each year of median annual earnings on estimated average schooling years, log population and median age, with heteroskedasticity-robust (HC3) standard errors.

1. The Question and the Data

Educational attainment and earnings differ across states, and examining their relationship provides a practical way to describe regional economic differences. This question is relevant to students, workforce analysts, and readers interested in how a state's educational composition relates to its earnings distribution. The goal is to measure this association carefully, rather than infer what would happen if a particular individual or state increased schooling.

The U.S. Census Bureau uses the American Community Survey to provide demographic, social, economic, and housing estimates. I use the 2022, 2023, and 2024 ACS 1-Year Estimates Detailed Tables, downloaded from data.census.gov. The public repository includes the original

CSV files, which the Colab notebook reads directly from GitHub. Keeping the original files makes the analysis reproducible without requiring readers to repeat the website filters.

The outcome is the overall Total estimate in B20004: median earnings over the past 12 months for people aged 25 and over with earnings. This is an annual earnings measure, not an hourly wage, household income, or earnings measure for a single education group. It excludes adults without earnings and does not limit the population to full-time, year-round workers, so differences in work intensity may contribute to variation across states.

B15003 reports educational attainment counts for all adults aged 25 and over. I convert these counts into an estimated average number of schooling years. No schooling, nursery school and kindergarten receive zero years; grades 1–11 receive their grade number. Grade 12 without a diploma, regular high school diploma and GED receive 12. Some college under one year receives 12.5; some college of one or more years without a degree receives 13.5. Associate's, bachelor's, master's, professional and doctorate degrees receive 14, 16, 18, 19 and 21 years, respectively.

For each state, I multiply the population in each attainment category by the years of schooling assigned to it, sum the results, and divide by the total population aged 25 and over. These assignments estimate completed schooling rather than measure actual attendance. For example, doctoral programs vary in length, and attainment categories do not capture every student's path. As a result, the estimated state average depends on the coding assumptions.

The controls are total population from B01003, entered as its natural logarithm, and overall median age from B01002. Both cover the entire state population, unlike the schooling and earnings measures. Statewide median age is a broad measure of age structure, not the median age specifically of adults with earnings.

The original downloads place states in columns. I retain the Estimate columns, reshape them so each state has its own row, remove commas before converting values to numbers, and exclude national totals, DC, and Puerto Rico. I check education category totals, then merge the four datasets by state and year with one-to-one validation. The final dataset contains all 50 states for each year, with no duplicates or missing model values.

No state-year observations are dropped because of missing model variables. Complete aggregate estimates do not show that the underlying survey was free of nonresponse or measurement error. The files do not indicate whether survey-level nonresponse is MCAR, MAR, or MNAR, so assigning any of these mechanisms would be unsupported. I distinguish the absence of missing values in my final table from the completeness of the underlying measurement process.

Table 1. Summary statistics

Year	Variable	Mean	Median	SD	Min	Max	N
2022	earnings	49119.24	48499.50	5534.19	40342.00	62009.00	50
	log_population	15.23	15.33	1.02	13.27	17.48	50
	median_age	39.33	39.15	2.15	32.10	45.10	50
	population	6652315.18	4551275.50	7451008.03	581381.00	39029342.00	50
	schooling_years	13.77	13.78	0.31	13.21	14.46	50
2023	earnings	50883.44	50132.50	5548.30	41175.00	65055.00	50
	log_population	15.23	15.33	1.02	13.28	17.48	50
	median_age	39.54	39.35	2.13	32.30	44.90	50
	population	6684718.48	4549951.50	7488403.38	584057.00	38965193.00	50
	schooling_years	13.79	13.80	0.31	13.19	14.41	50
2024	earnings	53173.48	51441.00	5429.95	44889.00	66968.00	50
	log_population	15.24	15.34	1.02	13.28	17.49	50
	median_age	39.62	39.40	2.12	32.50	44.90	50
	population	6788174.80	4593056.00	7630882.69	587618.00	39431263.00	50
	schooling_years	13.83	13.85	0.30	13.24	14.44	50

Note: Each state is weighted equally. Earnings are reported in each year's dollars; schooling and age are measured in years; population is measured in persons; and log population is its natural logarithm. Source: U.S. Census Bureau, ACS 1-Year Estimates, Tables B15003, B20004, B01003 and B01002.

Table 1 summarizes variation across states rather than individuals. In 2024, median state earnings are $53,173.48, while the median of the state-level earnings medians is $51,441. Earnings range from $44,889 to $66,968. These figures do not estimate the national mean or median, since small and large states receive equal weight.

Estimated average schooling varies less than earnings: in 2024, the state mean is 13.83 years, the standard deviation is about 0.30 years, and the range is approximately 13.24–14.44 years. Accordingly, the results section interprets both a 0.1-year difference and the coefficient for a full year. Population varies widely, motivating the use of logged state size rather than assuming that each additional resident has the same association in every state.

2. Descriptive Evidence

The figures show the data before controlling for other variables. Each panel uses the same axis scales across years, making visual comparisons consistent. They show the distribution of state-level estimates, not individual earnings.

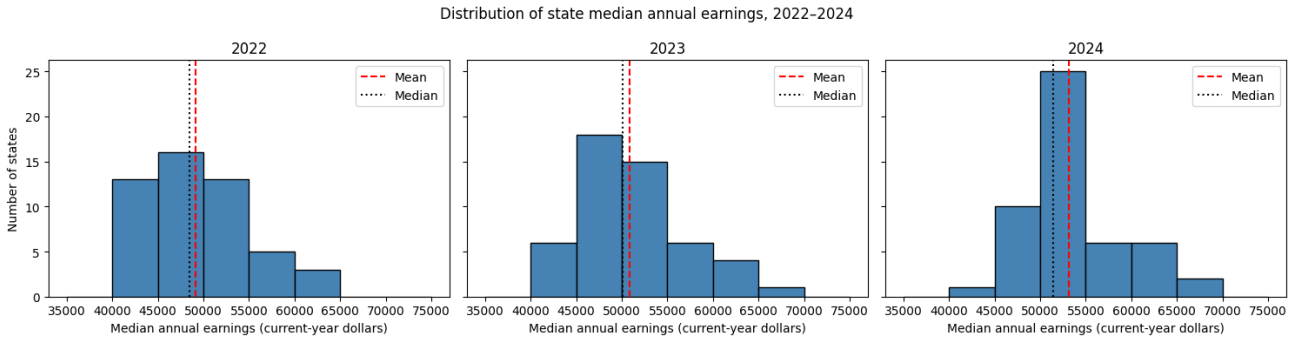


Figure 1. State median earnings are right-skewed in every year, with means above medians. Each panel contains 50 states. Red dashed lines mark means, while black dotted lines mark medians. Earnings are reported in current-year dollars, so year-to-year shifts alone do not establish changes in purchasing power. Source: ACS 1-Year Estimates, B20004, 2022–2024.

In 2022 and 2023, state medians cluster in the lower and middle earnings intervals, with fewer states in the upper intervals. The 2024 distribution clusters around $50,000–$55,000 and also has a right tail. Earnings remain in dollar levels so the regression coefficient can be interpreted as a dollar difference. OLS does not require a normally distributed outcome; residual behavior is examined separately.

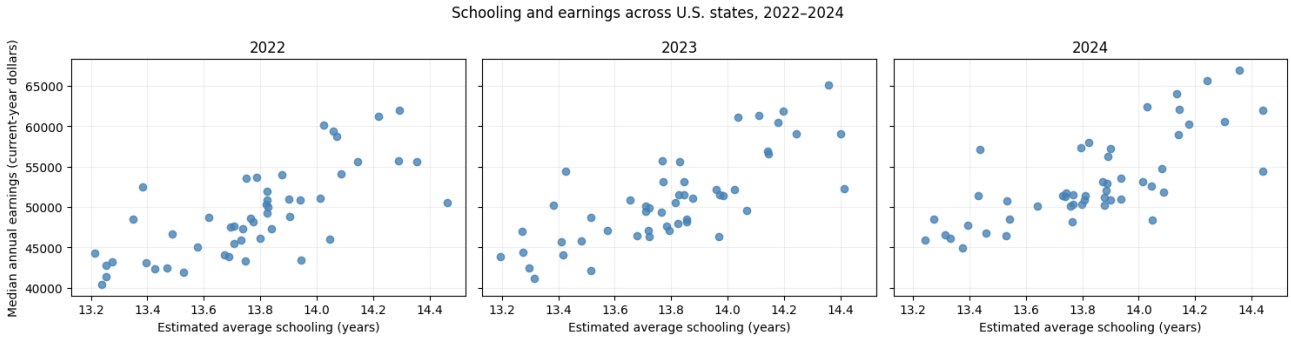


Figure 2. Higher estimated average schooling is associated with higher median annual earnings across all three years. Each point represents one state. These are unadjusted associations, without population or age controls, and do not establish causality. Source: ACS 1-Year Estimates, B15003 and B20004, 2022–2024.

The upward pattern appears in each annual panel, although states with similar estimated schooling still differ in earnings. The scatterplots therefore motivate regression but do not demonstrate that schooling alone accounts for the differences. Section 3 asks how the fitted association changes after accounting for state size and age structure.

3. The Model

For each year t, I estimate the following model across the 50 states s:
$Earnings_{st} = \alpha_t + \beta_t Schooling_{st} + \gamma_t \ln(Population_{st}) + \delta_t MedianAge_{st} + \varepsilon_{st}$.

I also estimate an uncontrolled model with only an intercept and schooling. The six estimates are reported in Table A1, with equal weight assigned to each state. I estimate separate regressions for each year; I do not combine the 150 state-year observations in one model or estimate state fixed effects. Thus, the results describe annual differences across states, not how changes in schooling within a state predict changes in earnings.

Schooling is the main explanatory variable. I include median age as a potential confounder because age structure may be related to both educational composition and earnings. This is a broad adjustment: a single median cannot capture the full age distribution or workers' experience. Log population adjusts for differences in state size, but does not directly measure urbanization, density, or industrial structure. Its inclusion alone also does not establish that a causal confounding path has been blocked.

The data cannot establish whether either control causes schooling or results from it. A variable affected by education may lie on a pathway through which education is associated with earnings; conditioning on a collider could also introduce bias. I therefore compare models with and without the controls and interpret both as associations. Neither a higher R-squared nor a statistically precise coefficient shows that the controls identify a causal effect.

All models use HC3 heteroskedasticity-robust standard errors. The displayed 95% confidence intervals use the normal reference distribution applied by the fitted models. HC3 allows error variance to differ across states but does not correct for omitted-variable bias, measurement error, or dependence between neighboring states. Because the sample includes all 50 states, the intervals are model-based measures of precision, not measures of uncertainty from randomly sampling 50 states from a larger set of U.S. states.

The controlled schooling coefficients are $14,814.51 in 2022, $14,999.86 in 2023, and $14,663.56 in 2024 for each one-year difference in estimated state average schooling. Their respective R-squared values are 0.633, 0.681, and 0.660. These values describe the fit within each annual sample; they do not measure education's causal contribution to earnings differences.

In 2024, a 0.1-year increase in estimated state average schooling is associated with approximately $1,466 higher median annual earnings, holding population and median age constant. The 95% confidence interval is approximately $1,114–$1,818. This is a comparison between state averages, not between two otherwise identical individuals with different lengths of schooling.

For population, a 10% larger is associated with approximately $150 higher median annual earnings, holding schooling and median age constant (95% CI: $37–$262). Since population is measured in logs while earnings remain in dollars, this estimate is calculated by multiplying the population coefficient ($1,568.84) by ln (1.10). It does not represent a percentage increase in earnings.

A one-year increase in a state's median age is associated with approximately $275 lower median annual earnings in 2024. The 95% interval ranges from $872 lower to $321 higher. The data are therefore consistent with associations in either direction, so the estimate should not be presented as clear evidence that older states have lower earnings.

Adding the controls raises the schooling coefficient relative to the uncontrolled estimates of $13,309.38, $13,902.47, and $13,309.49. The positive association persists after adjustment. The similar annual coefficients

suggest a recurring pattern, but because the same states appear each year, the three estimates are not independent replications. Amounts are reported in each year's dollars, so differences across annual coefficients do not test for changes in real returns.

4. Diagnostics

I assess the controlled OLS model separately for 2022, 2023, and 2024, using 50 states for each year. Each model includes estimated average schooling, log population, and median age. Following Lab 14, I examine residual plots, test for heteroskedasticity and residual non-normality, calculate variance inflation factors, and investigate influential observations. These checks assess the fitted models but cannot establish that the schooling coefficient is causal.

Residual plots

Figure A1 plots residuals against fitted earnings and estimated average schooling for each year. A residual is the difference between a state's observed median earnings and its fitted value. Positive residuals indicate that the model underpredicts earnings, while negative residuals mean it overpredicts them.

The plots show several large positive and negative deviations, as well as some clustering below zero in the middle ranges. No consistent funnel-shaped pattern appears across the three years. Still, these plots do not establish that the error variance is constant or that the linear specification is correct. The clustering suggests that the model may not capture all features of the conditional relationship. HC3 standard errors address heteroskedasticity but do not correct a misspecified conditional mean.

Heteroskedasticity and residual normality

The Breusch–Pagan test evaluates the null hypothesis that the error variance is constant. Its p-values are 0.2192 in 2022, 0.2961 in 2023, and 0.2082 in 2024. All exceed 0.05, so I do not reject the constant-variance hypothesis at the 5% level for any year. This does not prove that the errors are homoskedastic, particularly with only 50 observations per model. I therefore retain HC3 robust standard errors across all annual regressions.

The Jarque–Bera test evaluates residual normality based on skewness and kurtosis. Its p-values are 0.3783 in 2022, 0.3441 in 2023, and 0.2900 in 2024. These tests likewise do not reject the null hypothesis at the 5% level, providing no strong evidence against residual normality. However, failure to reject does not establish normality, and neither test addresses omitted-variable bias or reverse causality.

Multicollinearity

I calculate VIFs using each controlled model's design matrix, including its intercept, but report only the predictor VIFs. For schooling, log population and median age, respectively, the values are:

- 2022: 1.09, 1.04 and 1.07.
- 2023: 1.07, 1.04 and 1.06.
- 2024: 1.07, 1.06 and 1.07.

All values are close to one, indicating little variance inflation from linear relationships among the predictors. Multicollinearity therefore does not appear to be a major source of imprecision in these specifications. This

finding does not demonstrate that the controls are causally appropriate or that important variables have not been omitted.

Influential observations and sensitivity analysis

Following Lab 14, I examine leverage, standardized residuals, and Cook's distance. Leverage indicates how unusual a state's predictor combination is relative to those of other states. A standardized residual measures prediction error relative to its estimated standard deviation, while Cook's distance combines these measures to assess how much an observation influences the fitted model.

Utah has the largest Cook's distance in each of the three years: 0.2109 in 2022, 0.1525 in 2023, and 0.2231 in 2024. Its leverage values are 0.3092, 0.3076, and 0.3058, respectively, and its standardized residuals are −1.3730, −1.1719, and −1.4232. Utah's influence therefore reflects both its unusual predictor combination and its negative residual; its standardized residual is not the most extreme among the reported states. These diagnostics identify sensitivity, not a data error.

As a sensitivity check, I temporarily exclude the state with the largest Cook's distance in each year—Utah in all three cases—and refit the same controlled specification with HC3 standard errors. Table A2 reports the results:

2022: The schooling coefficient rises from $14,814.51 to $15,440.49, an increase of 4.23%. The revised 95% confidence interval is $11,309.09–$19,571.89.

2023: It rises from $14,999.86 to $15,449.76, an increase of 3.00%. The revised interval is $11,636.93–$19,262.59.

2024: It rises from $14,663.56 to $15,221.02, an increase of 3.80%. The revised interval is $11,434.12–$19,007.92.

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Each reduced sample includes 49 states. The coefficients remain positive, and all revised intervals remain above zero, suggesting that the positive association with schooling is not driven solely by Utah. I retain Utah in the main models because influence alone is not a reason to exclude a valid observation. This exclusion serves as a sensitivity check; it does not alter the main sample or provide evidence of causality.

5. Threats and Conclusion

Threats

For the schooling coefficient to have a causal interpretation, schooling would need to be unrelated to unobserved determinants of earnings after controlling for population size and median age. Reverse causality would also need to be absent, and the model's functional form and schooling measure would need to represent the intended relationship adequately.

These conditions are not established. Industry composition, regional prices and other omitted state characteristics may be related to both schooling and earnings. Higher earnings opportunities may also attract educated workers. These are possible explanations rather than mechanisms tested in this project. Passing the diagnostic tests does not resolve these identification problems.

Measurement creates further limitations. Schooling years are estimated from attainment categories using assigned values, rather than directly observed years attended. Schooling covers all adults aged 25+, whereas earnings cover adults aged 25+ with earnings. Median age and population cover the entire state population. Annual earnings also reflect differences in work intensity and are not adjusted for regional cost of living.

The main coefficient therefore represents a state-level conditional association: the difference in median earnings associated with higher estimated average schooling, holding population size and median age constant. It is not the income gain an individual would receive from another year of education.

Conclusion

Higher estimated average schooling is positively associated with median annual earnings in all three years. In 2024, a 0.1-year higher estimated state average schooling level is associated with approximately $1,466 higher median annual earnings, controlling for population size and median age (95% CI: $1,114–$1,818). Temporarily excluding Utah changes the annual schooling coefficients by 3.00%–4.23% without reversing their direction.

The recurring positive estimates and sensitivity check support confidence in the direction of the fitted association. However, they do not establish causality. Because the same states appear each year, the three annual estimates are not independent replications.

With another month, I would test different values for the years of schooling assigned to each education category, add theoretically justified measures of industry composition, and adjust earnings for regional price differences. These extensions would address specific measurement and comparability concerns, although they would not by themselves guarantee causal identification.

Appendix

Table A1. Annual OLS regressions of state median earnings on schooling and controls

	2022 Uncontrolled	2022 Controlled	2023 Uncontrolled	2023 Controlled	2024 Uncontrolled	2024 Controlled
schooling_years	13,309.38	14,814.51	13,902.47	14,999.86	13,309.49	14,663.56
schooling_years (SE)	(2,175.91)	(1,906.63)	(2,040.58)	(1,788.38)	(2,045.36)	(1,795.31)
log_population		1,428.54		1,592.56		1,568.84
log_population (SE)		(598.35)		(576.37)		(601.04)
median_age		-368.18		-174.85		-275.49
median_age (SE)		(324.70)		(284.89)		(304.58)
Intercept	-134,187.39	-162,184.78	-140,870.03	-173,347.23	-130,884.09	-162,608.01
Intercept (SE)	(29,774.35)	(28,123.99)	(28,001.53)	(27,351.46)	(28,082.39)	(27,315.78)
N	50	50	50	50	50	50
R-squared	0.541	0.633	0.589	0.681	0.555	0.660

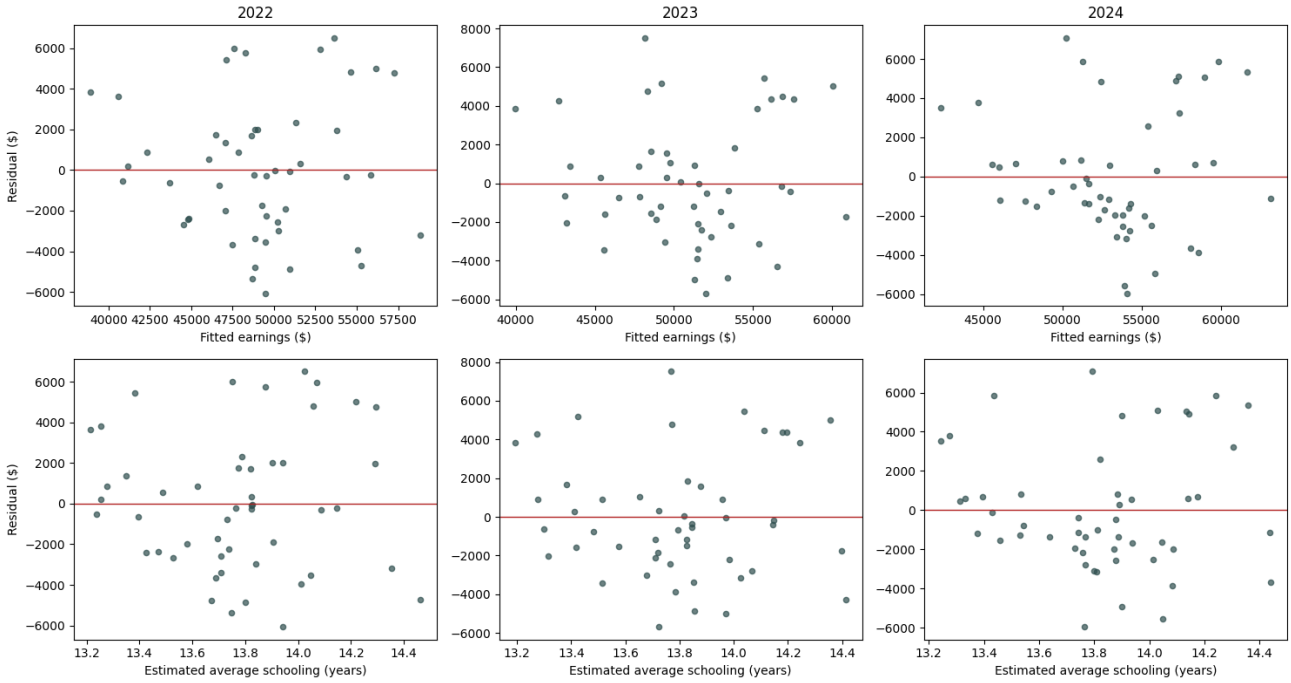
Note: The dependent variable is median annual earnings, measured in current-year dollars. HC3 robust standard errors are shown in parentheses. Each model includes all 50 states and an intercept. Uncontrolled models include schooling only; controlled models also include log population and median age. Source: Author's calculations using ACS 1-Year Estimates, 2022–2024.

Table A2. The positive schooling association persists after excluding Utah

	Year	Excluded state	Original coefficient	Revised coefficient	Change (%)	95% CI lower	95% CI upper	N
0	2022	Utah	14814.51	15440.49	4.23	11309.09	19571.89	49
1	2023	Utah	14999.86	15449.76	3.00	11636.93	19262.59	49
2	2024	Utah	14663.56	15221.02	3.80	11434.12	19007.92	49

Note: The controlled models are re-estimated without Utah, leaving 49 states in each year. Revised 95% confidence intervals are based on HC3 robust standard errors. Schooling coefficients are measured in dollars per estimated year of schooling. Change (%) is calculated relative to the original coefficient. Utah remains in the main sample. Source: Author's calculations using ACS 1-Year Estimates, 2022–2024.

Figure A1. Residuals and some clustering below zero in the middle ranges



Note: The top row shows residuals against fitted earnings; the bottom row shows residuals against estimated average schooling. Each column represents a separate year and includes all 50 states. Residuals and fitted earnings are measured in current-year dollars. Source: controlled OLS models reproduced in the notebook.